

Yu Sakurai

Ph.D. in Pharmaceutical Sciences

Targeted LNP, RNA delivery and ophthalmic development

serendipity100111@gmail.com

linkedin.com/in/yu-sakurai-5b679070 | Portfolio: yu-sakurai-research.yu1219.chatgpt.site

Senior Scientist specializing in targeted lipid nanoparticles (LNPs), with more than 10 years of academic RNA-delivery research and pharmaceutical R&D experience in Japan and Germany. Conceives and validates unexpected delivery routes, including ApoE-mediated uptake into lymphatic endothelial cells. Industry contributions include ophthalmic new-modality development, new excipient introduction and external technology due diligence.

Publication record: 54 original papers and 9 reviews | Verified September 2026

2022 Pharmaceutical Society of Japan Award for Young Scientists

Core expertise

Targeted RNA delivery: mRNA/siRNA-LNP systems; ligand and antibody modification; extrahepatic delivery; formulation design and optimization.

Quantitative research: DoE; Bayesian optimization; machine learning; Python and R; hands-on coding for proteomics and RNA-seq analysis.

Experimental development: LNP manufacturing methods; particle characterization and functional assays; in vitro and in vivo evaluation; omics-based toxicity analysis.

Professional experience

Boehringer Ingelheim Pharma GmbH & Co. KG | Germany

Senior Scientist, Global Development CMC | Jul 2025 - Present

Pharmaceutical R&D involving advanced drug-delivery technologies and emerging therapeutic concepts.

Lead equipment selection, technical alignment and procurement for new manufacturing technologies. Propose formulation concepts and provide technical/manufacturing support to other internal projects.

Boehringer Ingelheim Japan, Inc. | Japan

Scientist, Pharmaceutical Research | Dec 2023 - Jul 2025

Optimized formulations and manufacturing processes for new modalities, using machine learning and collaborating with analytical teams on formulation stability.

Initiated a new-modality proposal with research colleagues; coordinated NDA and MTA documentation with relevant internal specialists for external collaboration.

Contributions across pharmaceutical R&D roles

Ophthalmic development: evaluated new excipients and introduced a new excipient into a project through internal legal/patent coordination and technical discussions with multiple external partners.

Modality assessment: contributed specialist input to early-stage discussions and external technology due diligence, including on-site visits as an evaluation-team member.

Academic experience

Tohoku University | Japan

Assistant Professor / Lecturer | Apr 2021 - Nov 2023

Graduate School of Pharmaceutical Sciences.

Chiba University | Japan

Project Assistant Professor | Apr 2018 - Mar 2021

Graduate School of Pharmaceutical Sciences

Hokkaido University | Japan

Project Assistant Professor | Apr 2013 - Mar 2018

Graduate School of Pharmaceutical Sciences

Academic research leadership and contributions

Principal Investigator, JSPS KAKENHI 20K20195 (2020-2022): ligand modification of nanoparticles through click chemistry and systematic evaluation.

Worked on targeted LNP-mediated RNA delivery and extrahepatic delivery, including academic in vivo cell engineering and in vivo CAR-T applications.

Established LNP manufacturing methods and improved formulation performance through the incorporation of functional molecules.

Developed methodologies for LNP characterization and functional evaluation, and investigated administration routes through in vitro and in vivo studies.

Applied DoE to formulation optimization; codes proteomics and RNA-seq analyses. Published contributions include software and visualization for proteomics (Pharmaceutics, 2022;14:1560), and formal analysis and data curation for RNA-seq (Sci Rep, 2024;14:10503).

Supervised graduate and international students and contributed to research team development.

Publications and patents

Research record spanning nucleic acid therapeutics, targeted LNP delivery and formulation. Co-first and co-corresponding author of a sentinel-node formulation study using a five-factor, 16-condition DoE (Pharmaceutics, 2021;13:1462).

54 original papers and 9 reviews in the July 2026 publication list. Bibliographic details verified on 24 September 2026.

Co-inventor, WO 2023/054242 A1: LNP delivery to lymphatic endothelial cells (published application). Additional public patent records are linked in the research portfolio.

Education and certification

Ph.D. in Pharmaceutical Sciences | Hokkaido University, Japan

Licensed Pharmacist | Mar 2008

Tools and collaboration

Python, R, Google Apps Script; advanced Microsoft Excel; Word and PowerPoint. Experience working across Japan and Germany, coordinating external collaborations, and communicating across scientific disciplines and cultures.